Youssef Abdelhamid

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EDUCATION

Bachelor of Science in Computer Science (Software Engineering Program) | Assiut University 2022-2026
GPA: 3.32 / 4.00

PROJECTS

AI-Powered Real Estate Platform (Easy Sakan) | Python, FastAPI, XGBoost, YOLOv8, Surya OCR, OpenCV May 2026

- Architected and deployed an end-to-end Machine Learning backend for a housing platform, bridging Python-based AI engines with a Next.js administrative dashboard via FastAPI.
- Engineered a 6-vector Fraud Detection Pipeline integrating custom-trained YOLOv8 (stamp detection), Surya OCR (text extraction), and OpenCV (ELA manipulation analysis) with a strict fail-closed security policy.
- Developed an XGBoost Pricing Engine to dynamically predict fair market rent, achieving an R^2 of 0.98 and an MAE of ±346 EGP, integrated into a real-time heuristic deal-rating UI.
- Implemented a sub-5ms Recommendation Engine utilizing Jaccard Similarity and custom weighted mathematical scoring to serve dynamic, session-based apartment recommendations.

Student Dropout Early Warning System | Python, XGBoost, Pandas, Streamlit April 2026

- Developed an early warning predictive model using XGBoost to identify at-risk students based on a 4.4K+ record dataset comprising academic and socio-economic features.
- Achieved 93.11% test accuracy, 0.97 ROC-AUC score, and 0.91 F1-Score, validated through 5-fold cross-validation.
- Built and deployed an interactive Streamlit dashboard for real-time risk assessment and dynamic feature mapping.

Real-Time Posture & Attention Monitor | Python, OpenCV, MediaPipe, NumPy April 2026

- Implemented a live computer vision monitoring system using MediaPipe (Face & Pose Landmarkers) to track user posture and gaze geometry (yaw/pitch).
- Applied Exponential Moving Average (EMA) to stabilize landmark coordinate metrics and eliminate real-time noisy readings.
- Integrated dynamic calibration and background segmentation utilizing TFLite models for real- time ergonomic feedback.

NLP Sentiment Classifier & Analyzer | Python, Logistic Regression, TF-IDF, VADER May 2025

- Built a 3-class sentiment analyzer trained on a balanced 31K review dataset, utilizing class_weight='balanced' to prevent majority-class bias.
- Augmented TF-IDF unigram/trigram vectorization with pre-computed VADER sentiment scores (neg, neu, pos, compound) to capture textual emphasis and punctuation signals.
- Applied sublinear TF scaling for frequency normalization, achieving a solid 68.3% accuracy across complex real-world text inputs.

TECHNICAL SKILLS

- **Programming Languages & Tools:** Python, C++, C, C#, JavaScript, SQL, Git, Linux, Docker, Postman
- **Frameworks & Libraries:** Flask, FastAPI, Scikit-learn, Pandas, Matplotlib, Seaborn, XGBoost, PyTorch, Keras, OpenCV
- **Languages:** English: IELTS 6.0 (B2 Equivalent), German: Basic, Arabic: Native